

Simple Nonlinear Graphs

Solutions



Curriculum Ready
ACMNA: 296, 239, 267



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Page 1 questions

1. Identify which of these relationships is linear.

a $y = x + 2$

Linear

b $y = x^2 + 2x$

Nonlinear

c $y = -3x + 3$

Linear

d $x + 2y = 6$

Linear

e $y = \frac{x}{2}$

Linear

f $y = \frac{2}{x}$

Nonlinear

2. Complete the tables for these equations.

a $y = x + 3$

x	-2	-1	0	1	2
y	1	2	3	4	5

b $y = 2x$

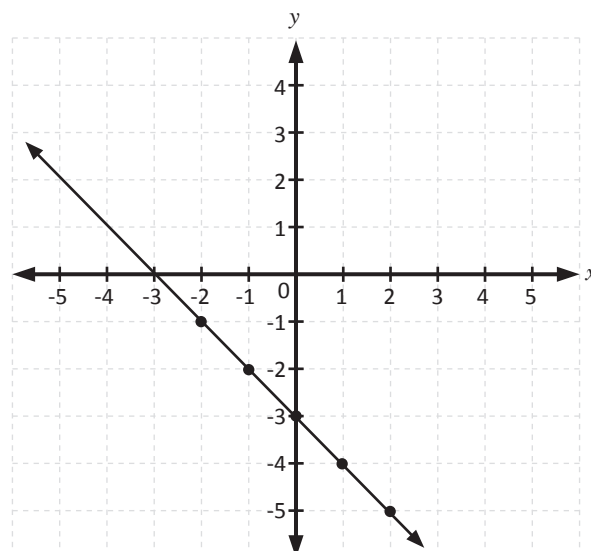
x	-2	-1	0	1	2
y	-4	-2	0	2	4

c $y = -x$

x	-2	-1	0	1	2
y	2	1	0	-1	-2

d $y = -x - 3$

x	-2	-1	0	1	2
y	-1	-2	-3	-4	-5

3. Use the table d in the previous question to plot the line $y = -x - 3$ on the coordinate plane below.

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1. Identify whether these relationships are linear or quadratic.

a $y = -x + 2$

Linear

b $y = x^2$

Quadratic

c $y = \frac{-3}{2}x^2$

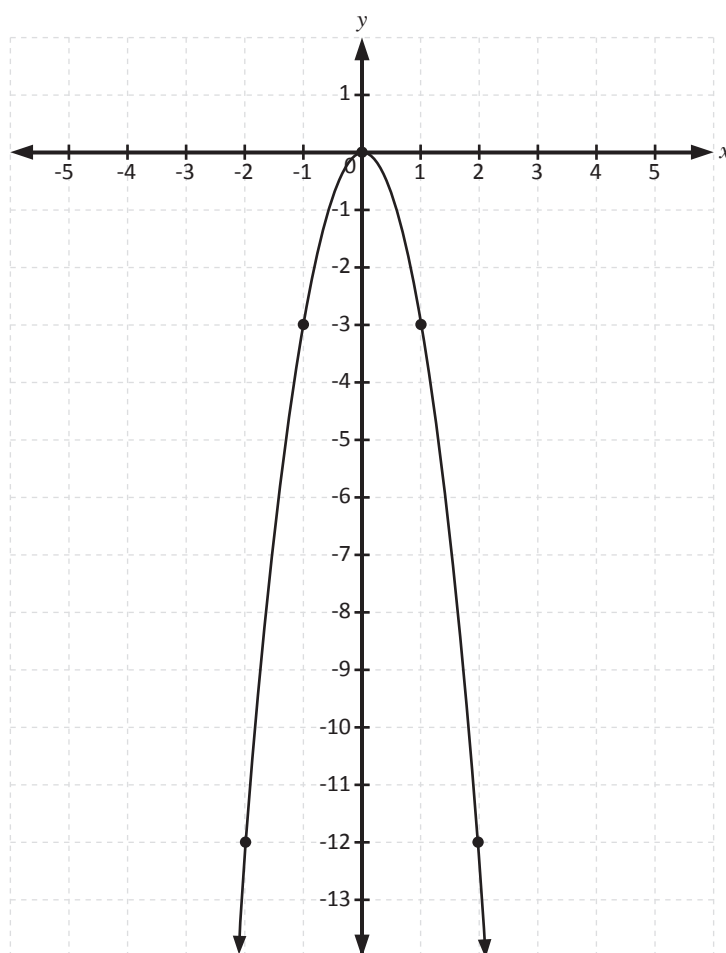
Quadratic

d $y = x + x^2$

Quadratic

2. Complete the table and draw the graph of $y = -3x^2$.

x	-2	-1	0	1	2
y	-12	-3	0	-3	-12



Is this parabola concave upwards or concave downwards?

Concave downward (sad)

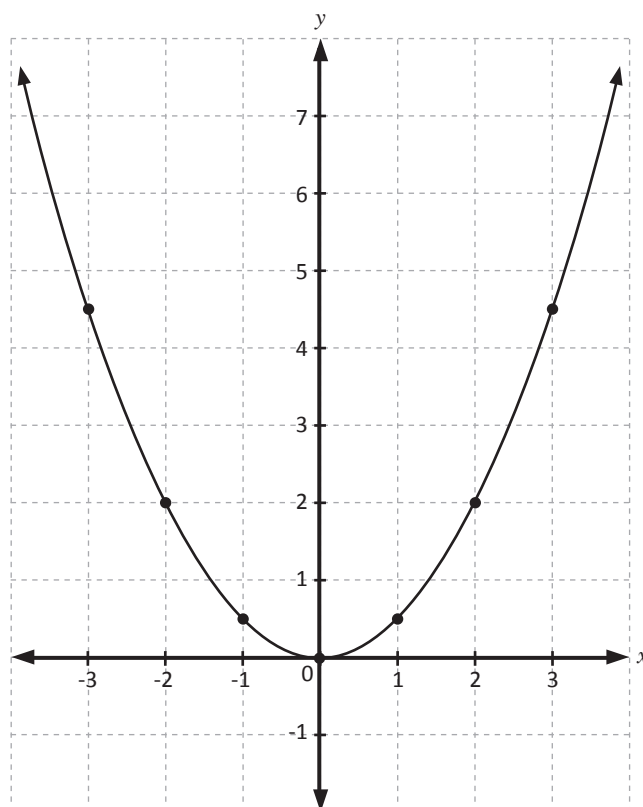
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3. Answer these questions based on the function $y = \frac{1}{2}x^2$.

- a Complete this table based on the function.

x	-3	-2	-1	0	1	2	3
y	4.5	2	0.5	0	0.5	2	4.5

- b Use the space below to draw your own set of axes and the graph of this function.



- c Is this parabola concave upwards or concave downwards?

Concave upwards (happy)

4. Identify if the parabola for these functions is concave upwards or concave downwards (without drawing the graphs).

a $y = -7x^2$

Concave downward (sad)

b $y = \frac{x^2}{3}$

Concave upwards (happy)

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5. $y = \frac{1}{4}x^2$ is a quadratic function.

- a Complete this table for this function.

x	-4	-2	0	2	4
y	4	1	0	1	4

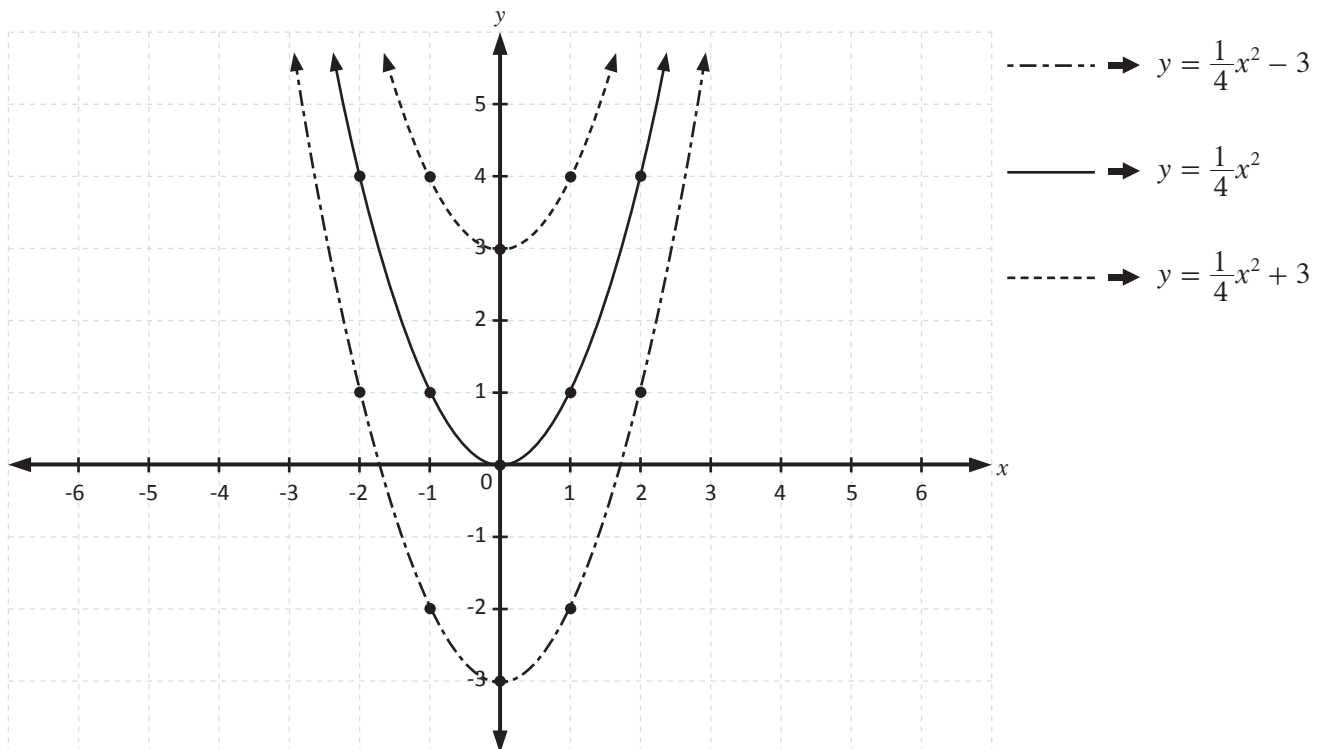
- b Complete the table for the function $y = \frac{1}{4}x^2 + 3$.

x	-4	-2	0	2	4
y	7	4	3	4	7

- c Complete the table for the function $y = \frac{1}{4}x^2 - 3$.

x	-4	-2	0	2	4
y	1	-2	-3	-2	1

- d Use the table to plot all 3 graphs on this system of axes.



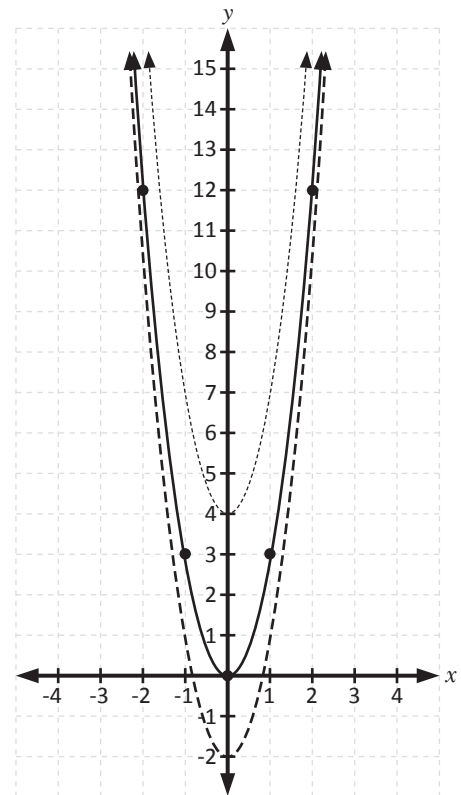
- e What do you notice about the graphs of $y = \frac{1}{4}x^2 + 3$ and $y = \frac{1}{4}x^2 - 3$ in relation to $y = \frac{1}{4}x^2$?

When 3 is added the curve moves 3 units up the y -axis. Similarly, when 3 is subtracted the curve moves 3 units down the y -axis.

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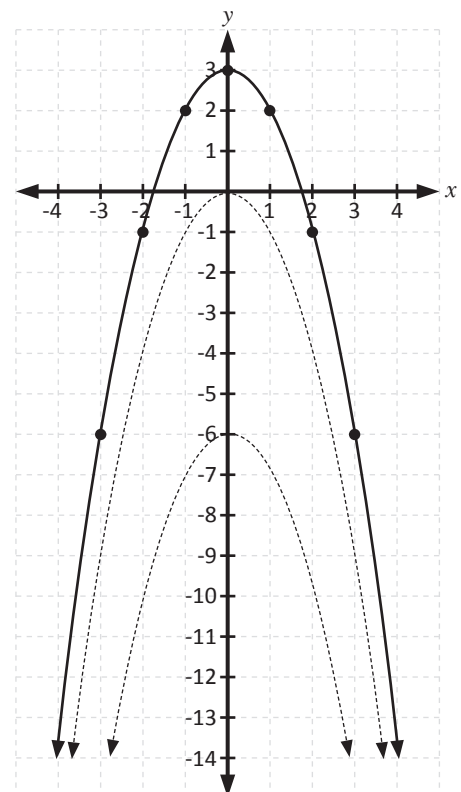
6. The graph on the right represents $y = 3x^2$.

- a On the same set of axes, draw the graph of $y = 3x^2 + 4$ without using a table.
- b On the same set of axes, draw the graph of $y = 3x^2 - 2$ without using a table.
- c Are these graphs concave upwards or concave downwards?
All three graphs are concave upwards (happy)
- d Does adding or subtracting a constant change the concavity?
No. Adding or subtracting a constant does not change concavity



7. The graph on the right represents $y = -x^2 + 3$.

- a On the same set of axes, draw the graph of $y = -x^2$ without using a table.
- b On the same set of axes, draw the graph of $y = -x^2 - 6$ without using a table.
- c Are these graphs concave upwards or concave downwards?
All three graphs are concave downwards (sad)
- d Does adding or subtracting a constant change the concavity?
No. Adding or subtracting a constant does not change concavity



Page 13 questions

1. Why do some parabolas have maximum turning points and others have minimum turning points?

Concave upward parabolas (where $a > 0$) have a turning point that is a minimum. While concave downward parabolas (where $a < 0$) have a turning point that is a maximum.

2. Use a table to draw this parabola: $y = 4 - x^2$ on the axes provided. (Hint: $y = 4 - x^2$ is the same as $y = -x^2 + 4$)

a

x	-2	-1	0	1	2
y	0	3	4	3	0

b What is the vertex of this parabola?

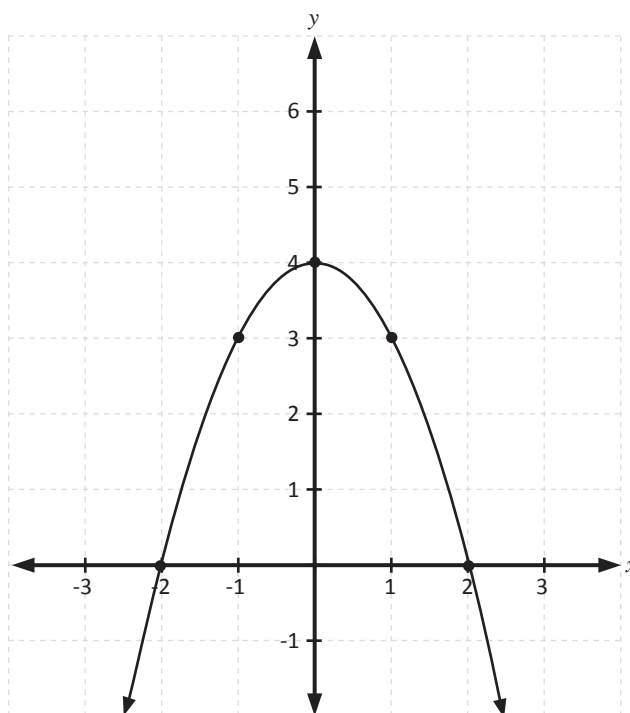
(0, 4)

c Is the vertex a maximum or a minimum?

Maximum. Concave downwards (sad) parabolas always have a vertex that is a maximum.

d What is the axis of symmetry?

$x = 0$



3. Without drawing anything, state whether each of the following will have a minimum or maximum turning point.

a $y = 2x^2 - 3$

$a > 0 \therefore$ turning point is a minimum.
Minimum at (0, -3)

b $y = -3x^2 + 7$

$a < 0 \therefore$ turning point is a maximum.
Maximum at (0, 7)

c $y = 5 + 6x^2$

$a > 0 \therefore$ turning point is a minimum.
Minimum at (0, 5)

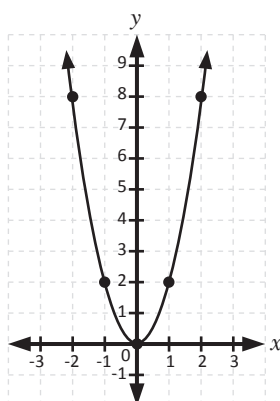
d $y = 5 - 5x^2$

$a < 0 \therefore$ turning point is a maximum.
Maximum at (0, 5)

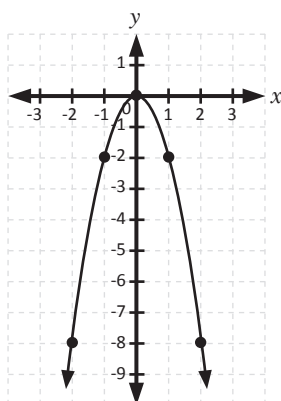
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4. Match these equations to the parabolas below.

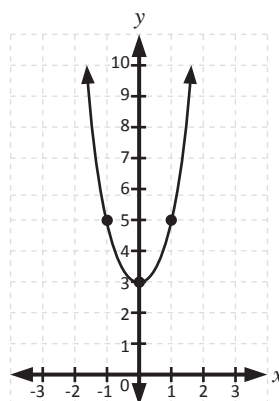
a $y = 2x^2$



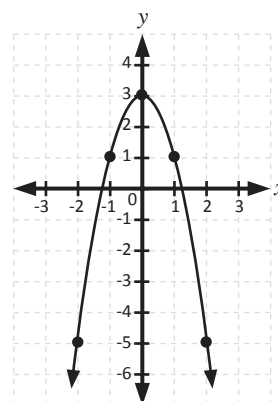
$y = -2x^2$



$y = 2x^2 + 3$



$y = 3 - 2x^2$

b Identify which equation has a maximum turning point at $(0,0)$.

$y = -2x^2$

c Identify which equation has a minimum turning point at $(0,0)$.

$y = 2x^2$

d What is the axis of symmetry of all these functions?

$x = 0$

e Identify which equation has a maximum y-value of 3.

$y = 3 - 2x^2$

f Identify which equation has a minimum y-value of 3.

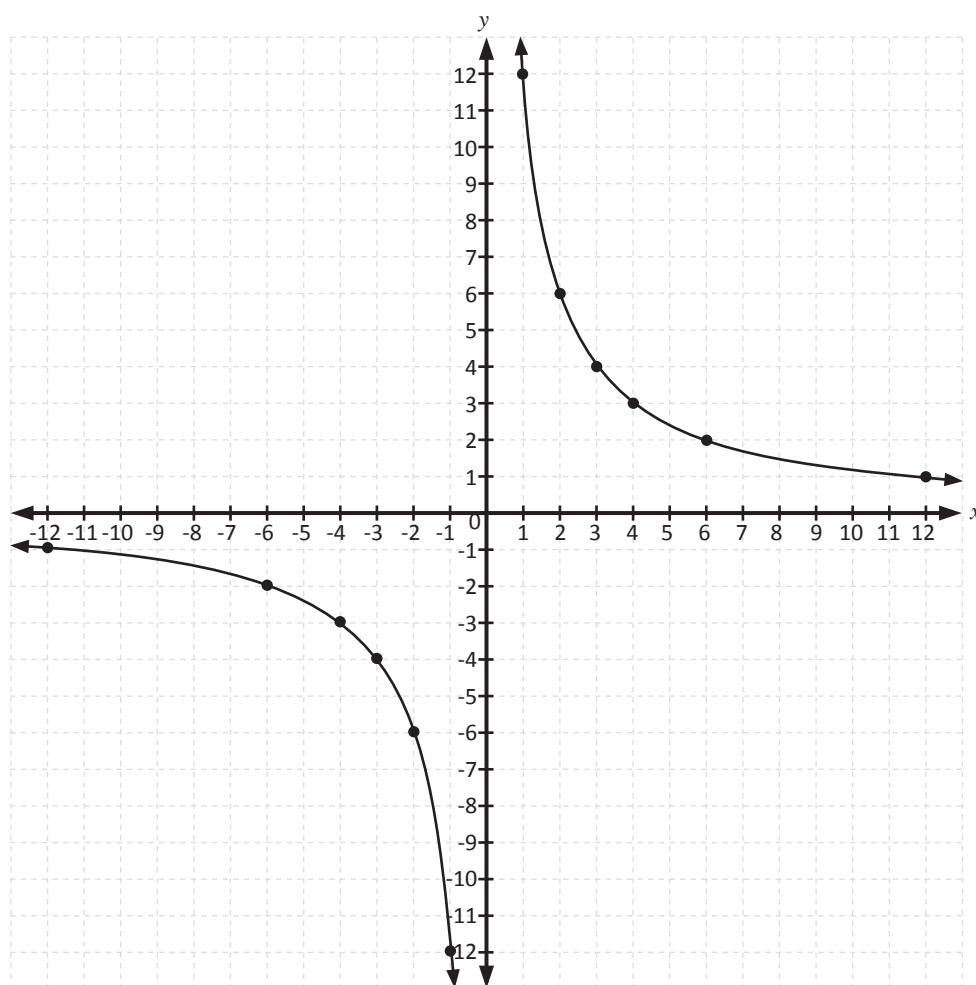
$y = 2x^2 + 3$

Page 18 questions

1. $y = \frac{12}{x}$ can be written as $xy = 12$.

- a Use the table and axes below to draw the graph of $xy = 12$.

x	-6	-4	-3	-2	-1	1	2	3	4	6
y	-2	-3	-4	-6	-12	12	6	4	3	2



- b Will either of the branches ever touch the x or y axis?

No. Both the x and y axes are asymptotes.

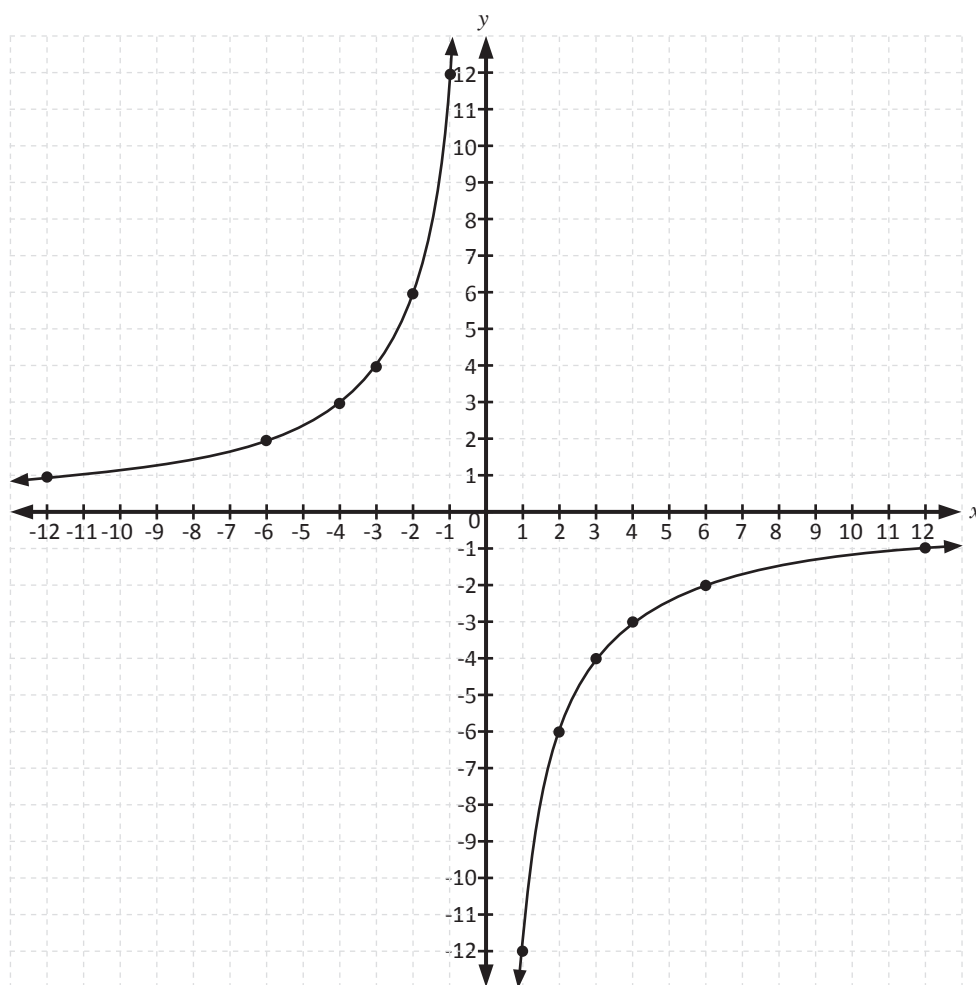
- c Which quadrants are the branches of the hyperbola in?

The hyperbola has a branch in each of the 1st and 3rd quadrants because $k > 0$.

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- d Use the table and axes below to draw the graph of $xy = -12$.

x	-6	-4	-3	-2	-1	1	2	3	4	6
y	2	3	4	6	12	-12	-6	-4	-3	-2



- e What is different in the above hyperbola from the hyperbola on the previous page?

This hyperbola has a branch in each of the 2nd and 4th quadrants (rather than 1st and 3rd)

- f Highlight the asymptotes on each of the above graphs.

The asymptotes for both of these hyperbolas is the x and y axes. This means that the graph will get very close to these asymptotes but will never touch these.

Page 20 questions

2. The exponential equation is an equation of the form $y = a^x$ where a is any positive number.

- a What is the value of 5^0 ?

The value of $5^0 = 1$

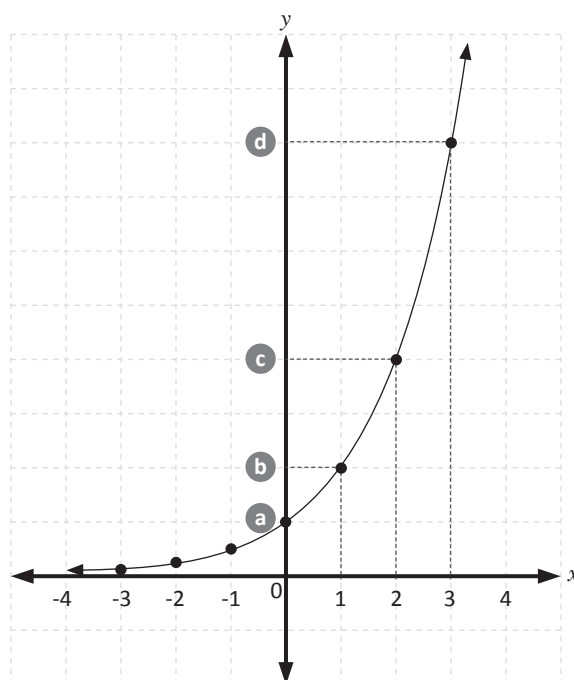
- b What is the value of a^0 where a is any positive number?

The value of $a^0 = 1$

- c What is the y-intercept of any exponential equation $y = a^x$? Why?

The y-intercept is always 1 for any exponential equation of the form $y = a^x$. This is because any number raised to the power of zero will be one.

3. Find the missing values on the y-axis, if the graph below is of the function $y = 4^x$.



- a $(0, 1) \therefore y = 1$ b $(1, 4) \therefore y = 4$ c $(2, 16) \therefore y = 16$ d $(3, 64) \therefore y = 64$

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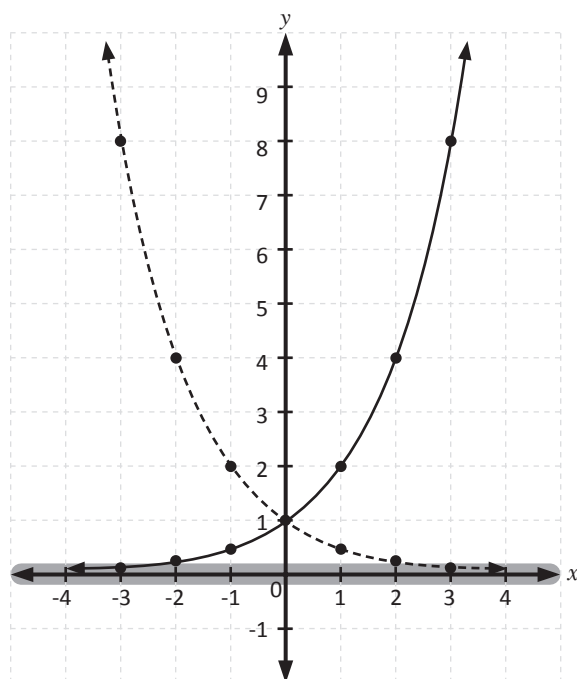
4. The table below has been completed for $y = 2^x$ and the graph plotted underneath.

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

a Complete the table below for $y = 2^{-x}$.

x	-3	-2	-1	0	1	2	3
y	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

b Plot the graph of $y = 2^{-x}$ on the same set of axes above.



c In the graph of $y = 2^x$, y increases as x increases. Is this the same in $y = 2^{-x}$?

No. As x increases, y decreases. The equation $y = 2^{-x}$ is shown in the above graph using the dashed line.

d Highlight the asymptotes for both graphs.

The asymptote for both graphs is the same. It is the x -axis or the line $y = 0$. In other words, while both the curves $y = 2^x$ and $y = 2^{-x}$ get very close to the x -axis, they never touch it.



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